

12.1-Support_Vector_Classifier

August 7, 2025

1 Support Vector Classifier Implementation

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
from sklearn.datasets import make_classification
```

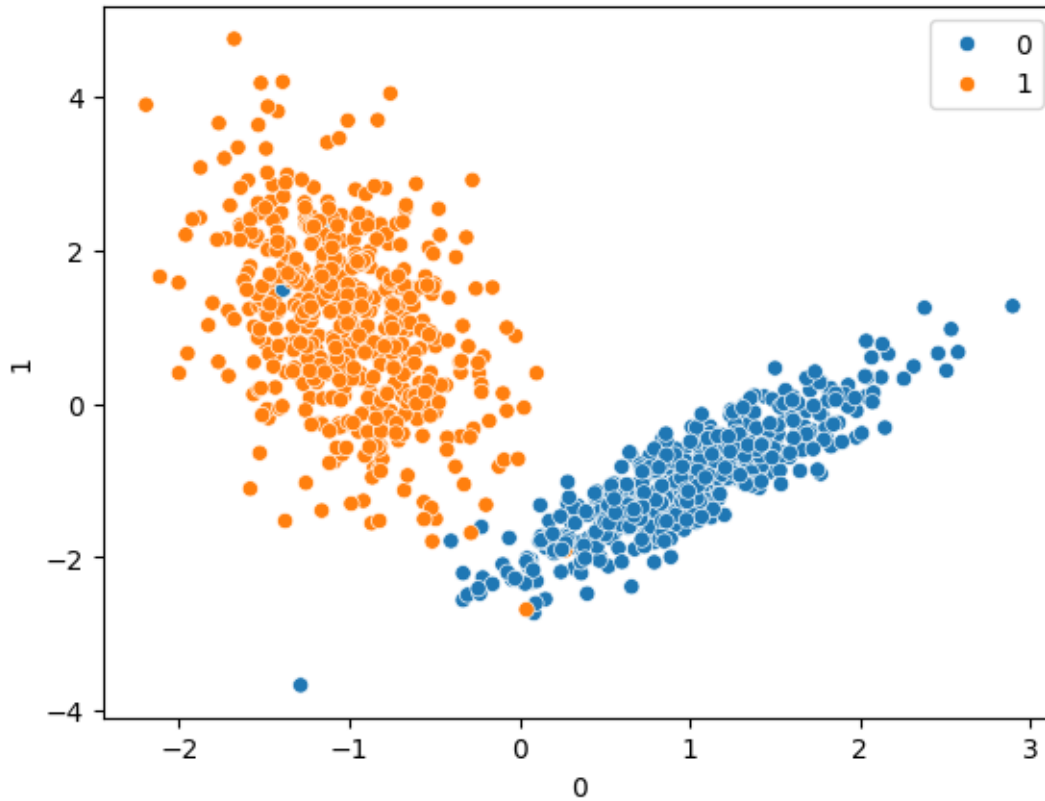
```
[2]: ## crating the synthetic data points
X, y = make_classification(n_samples=1000, n_features=2, n_classes=2,
↪n_clusters_per_class=1, n_redundant=0)
```

```
[3]: pd.DataFrame(X).head()
```

```
[3]:      0      1
0 -1.109463  1.768061
1 -0.853484 -0.830759
2  0.769077 -1.269778
3  0.359453 -1.797662
4 -1.473704 -0.190234
```

```
[4]: sns.scatterplot(x = pd.DataFrame(X)[0], y = pd.DataFrame(X)[1], hue=y)
```

```
[4]: <Axes: xlabel='0', ylabel='1'>
```



```
[5]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25,
↳random_state=23)
```

```
[6]: from sklearn.svm import SVC
svc = SVC(kernel='linear')
svc.fit(X_train, y_train)
```

```
[6]: SVC(kernel='linear')
```

```
[7]: y_pred = svc.predict(X_test)
```

```
[8]: from sklearn.metrics import accuracy_score, confusion_matrix,
↳classification_report
print(accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
```

0.992

	precision	recall	f1-score	support
0	0.99	0.99	0.99	133

1	0.99	0.99	0.99	117
accuracy			0.99	250
macro avg	0.99	0.99	0.99	250
weighted avg	0.99	0.99	0.99	250

```
[[132  1]
 [  1 116]]
```

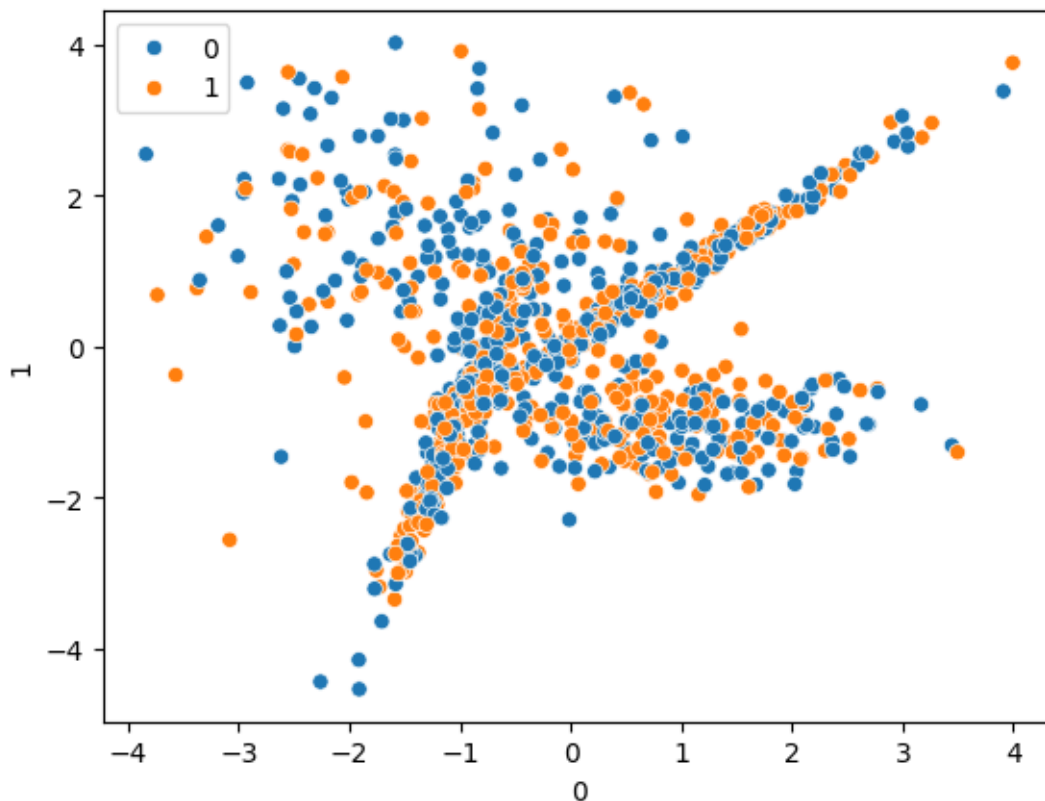
Due to clearly separable points we get the good accuracy

if we increase the clusters_pre_class is more then what happens lets see

```
[9]: ## Creating the data
X_new, y_new = make_classification(n_samples=1000, n_features=2, n_classes=2,
    ↪n_clusters_per_class=2, n_redundant=0, random_state=42)
```

```
[10]: sns.scatterplot(x = pd.DataFrame(X_new)[0], y = pd.DataFrame(X_new)[1], hue=y)
```

```
[10]: <Axes: xlabel='0', ylabel='1'>
```



```
[11]: X_new_train, X_new_test, y_new_train, y_new_test = train_test_split(X_new, y_new, test_size=0.3, random_state=42)
```

```
[12]: svc_new = SVC(kernel='linear')
svc_new.fit(X_new_train, y_new_train)
y_new_pred = svc_new.predict(X_new_test)
```

```
[13]: print(accuracy_score(y_new_test, y_new_pred))
print(classification_report(y_new_test, y_new_pred))
print(confusion_matrix(y_new_test, y_new_pred))
```

```
0.8766666666666667
              precision    recall  f1-score   support

    0           0.86       0.90       0.88        148
    1           0.90       0.86       0.88        152

 accuracy                   0.88        300
 macro avg              0.88       0.88       0.88        300
weighted avg              0.88       0.88       0.88        300
```

```
[[133  15]
 [ 22 130]]
```

Here you can see that the accuracy has dropped

1.1 Applying rbf Kernel

```
[14]: rbf = SVC(kernel='rbf')
rbf.fit(X_new_train, y_new_train)
```

```
[14]: SVC()
```

```
[15]: y_pred=rbf.predict(X_new_test)
```

```
[16]: print(accuracy_score(y_new_test, y_pred))
print(classification_report(y_new_test, y_pred))
```

```
0.8866666666666667
              precision    recall  f1-score   support

    0           0.85       0.94       0.89        148
    1           0.93       0.84       0.88        152

 accuracy                   0.89        300
 macro avg              0.89       0.89       0.89        300
weighted avg              0.89       0.89       0.89        300
```

2 Hyperparameter Tuning

```
[17]: from sklearn.model_selection import GridSearchCV
```

```
params = {'C': [0.1, 1, 10, 100, 1000],  
          'gamma': [1, 0.1, 0.01, 0.001, 0.0001],  
          'kernel': ['rbf']}
```

```
[18]: grid = GridSearchCV(SVC(), param_grid=params, refit=True, cv=5, verbose=3 )
```

```
[19]: grid.fit(X_new_train, y_new_train)
```

Fitting 5 folds for each of 25 candidates, totalling 125 fits

```
[CV 1/5] END ..C=0.1, gamma=1, kernel=rbf;; score=0.921 total time= 0.0s  
[CV 2/5] END ..C=0.1, gamma=1, kernel=rbf;; score=0.864 total time= 0.0s  
[CV 3/5] END ..C=0.1, gamma=1, kernel=rbf;; score=0.886 total time= 0.0s  
[CV 4/5] END ..C=0.1, gamma=1, kernel=rbf;; score=0.850 total time= 0.0s  
[CV 5/5] END ..C=0.1, gamma=1, kernel=rbf;; score=0.864 total time= 0.0s  
[CV 1/5] END ..C=0.1, gamma=0.1, kernel=rbf;; score=0.886 total time= 0.0s  
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[CV 3/5] END ..C=0.1, gamma=0.1, kernel=rbf;; score=0.879 total time= 0.0s  
[CV 4/5] END ..C=0.1, gamma=0.1, kernel=rbf;; score=0.864 total time= 0.0s  
[CV 5/5] END ..C=0.1, gamma=0.1, kernel=rbf;; score=0.821 total time= 0.0s  
[CV 1/5] END ..C=0.1, gamma=0.01, kernel=rbf;; score=0.886 total time= 0.0s  
[CV 2/5] END ..C=0.1, gamma=0.01, kernel=rbf;; score=0.857 total time= 0.0s  
[CV 3/5] END ..C=0.1, gamma=0.01, kernel=rbf;; score=0.879 total time= 0.0s  
[CV 4/5] END ..C=0.1, gamma=0.01, kernel=rbf;; score=0.850 total time= 0.0s  
[CV 5/5] END ..C=0.1, gamma=0.01, kernel=rbf;; score=0.793 total time= 0.0s  
[CV 1/5] END ..C=0.1, gamma=0.001, kernel=rbf;; score=0.886 total time= 0.0s  
[CV 2/5] END ..C=0.1, gamma=0.001, kernel=rbf;; score=0.857 total time= 0.0s  
[CV 3/5] END ..C=0.1, gamma=0.001, kernel=rbf;; score=0.886 total time= 0.0s  
[CV 4/5] END ..C=0.1, gamma=0.001, kernel=rbf;; score=0.886 total time= 0.0s  
[CV 5/5] END ..C=0.1, gamma=0.001, kernel=rbf;; score=0.829 total time= 0.0s  
[CV 1/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=0.886 total time= 0.0s  
[CV 2/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=0.850 total time= 0.0s  
[CV 3/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=0.886 total time= 0.0s  
[CV 4/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=0.886 total time= 0.0s  
[CV 5/5] END ..C=0.1, gamma=0.0001, kernel=rbf;; score=0.829 total time= 0.0s  
[CV 1/5] END ..C=1, gamma=1, kernel=rbf;; score=0.921 total time= 0.0s  
[CV 2/5] END ..C=1, gamma=1, kernel=rbf;; score=0.893 total time= 0.0s  
[CV 3/5] END ..C=1, gamma=1, kernel=rbf;; score=0.900 total time= 0.0s  
[CV 4/5] END ..C=1, gamma=1, kernel=rbf;; score=0.871 total time= 0.0s  
[CV 5/5] END ..C=1, gamma=1, kernel=rbf;; score=0.864 total time= 0.0s  
[CV 1/5] END ..C=1, gamma=0.1, kernel=rbf;; score=0.900 total time= 0.0s  
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[CV 5/5] END ..C=1, gamma=0.1, kernel=rbf;; score=0.821 total time= 0.0s
```

[CV 1/5]	END	...C=1, gamma=0.01, kernel=rbf;	score=0.871 total time=	0.0s
[CV 2/5]	END	...C=1, gamma=0.01, kernel=rbf;	score=0.850 total time=	0.0s
[CV 3/5]	END	...C=1, gamma=0.01, kernel=rbf;	score=0.893 total time=	0.0s
[CV 4/5]	END	...C=1, gamma=0.01, kernel=rbf;	score=0.879 total time=	0.0s
[CV 5/5]	END	...C=1, gamma=0.01, kernel=rbf;	score=0.836 total time=	0.0s
[CV 1/5]	END	...C=1, gamma=0.001, kernel=rbf;	score=0.886 total time=	0.0s
[CV 2/5]	END	...C=1, gamma=0.001, kernel=rbf;	score=0.857 total time=	0.0s
[CV 3/5]	END	...C=1, gamma=0.001, kernel=rbf;	score=0.879 total time=	0.0s
[CV 4/5]	END	...C=1, gamma=0.001, kernel=rbf;	score=0.850 total time=	0.0s
[CV 5/5]	END	...C=1, gamma=0.001, kernel=rbf;	score=0.793 total time=	0.0s
[CV 1/5]	END	...C=1, gamma=0.0001, kernel=rbf;	score=0.886 total time=	0.0s
[CV 2/5]	END	...C=1, gamma=0.0001, kernel=rbf;	score=0.850 total time=	0.0s
[CV 3/5]	END	...C=1, gamma=0.0001, kernel=rbf;	score=0.886 total time=	0.0s
[CV 4/5]	END	...C=1, gamma=0.0001, kernel=rbf;	score=0.886 total time=	0.0s
[CV 5/5]	END	...C=1, gamma=0.0001, kernel=rbf;	score=0.829 total time=	0.0s
[CV 1/5]	END	...C=10, gamma=1, kernel=rbf;	score=0.914 total time=	0.0s
[CV 2/5]	END	...C=10, gamma=1, kernel=rbf;	score=0.921 total time=	0.0s
[CV 3/5]	END	...C=10, gamma=1, kernel=rbf;	score=0.900 total time=	0.0s
[CV 4/5]	END	...C=10, gamma=1, kernel=rbf;	score=0.886 total time=	0.0s
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[CV 1/5]	END	...C=10, gamma=0.1, kernel=rbf;	score=0.914 total time=	0.0s
[CV 2/5]	END	...C=10, gamma=0.1, kernel=rbf;	score=0.879 total time=	0.0s
[CV 3/5]	END	...C=10, gamma=0.1, kernel=rbf;	score=0.886 total time=	0.0s
[CV 4/5]	END	...C=10, gamma=0.1, kernel=rbf;	score=0.864 total time=	0.0s
[CV 5/5]	END	...C=10, gamma=0.1, kernel=rbf;	score=0.850 total time=	0.0s
[CV 1/5]	END	...C=10, gamma=0.01, kernel=rbf;	score=0.879 total time=	0.0s
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[CV 3/5]	END	...C=10, gamma=0.01, kernel=rbf;	score=0.893 total time=	0.0s
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[CV 1/5]	END	...C=10, gamma=0.0001, kernel=rbf;	score=0.886 total time=	0.0s
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[CV 3/5]	END	...C=10, gamma=0.0001, kernel=rbf;	score=0.879 total time=	0.0s
[CV 4/5]	END	...C=10, gamma=0.0001, kernel=rbf;	score=0.850 total time=	0.0s
[CV 5/5]	END	...C=10, gamma=0.0001, kernel=rbf;	score=0.793 total time=	0.0s
[CV 1/5]	END	...C=100, gamma=1, kernel=rbf;	score=0.900 total time=	0.0s
[CV 2/5]	END	...C=100, gamma=1, kernel=rbf;	score=0.921 total time=	0.0s
[CV 3/5]	END	...C=100, gamma=1, kernel=rbf;	score=0.900 total time=	0.0s
[CV 4/5]	END	...C=100, gamma=1, kernel=rbf;	score=0.879 total time=	0.0s
[CV 5/5]	END	...C=100, gamma=1, kernel=rbf;	score=0.850 total time=	0.0s
[CV 1/5]	END	...C=100, gamma=0.1, kernel=rbf;	score=0.929 total time=	0.0s
[CV 2/5]	END	...C=100, gamma=0.1, kernel=rbf;	score=0.886 total time=	0.0s
[CV 3/5]	END	...C=100, gamma=0.1, kernel=rbf;	score=0.893 total time=	0.0s

```

[CV 4/5] END ...C=100, gamma=0.1, kernel=rbf;, score=0.864 total time= 0.0s
[CV 5/5] END ...C=100, gamma=0.1, kernel=rbf;, score=0.871 total time= 0.0s
[CV 1/5] END ...C=100, gamma=0.01, kernel=rbf;, score=0.886 total time= 0.0s
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[CV 5/5] END ...C=100, gamma=0.01, kernel=rbf;, score=0.814 total time= 0.0s
[CV 1/5] END ...C=100, gamma=0.001, kernel=rbf;, score=0.879 total time= 0.0s
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[CV 1/5] END ...C=100, gamma=0.0001, kernel=rbf;, score=0.871 total time= 0.0s
[CV 2/5] END ...C=100, gamma=0.0001, kernel=rbf;, score=0.836 total time= 0.0s
[CV 3/5] END ...C=100, gamma=0.0001, kernel=rbf;, score=0.886 total time= 0.0s
[CV 4/5] END ...C=100, gamma=0.0001, kernel=rbf;, score=0.879 total time= 0.0s
[CV 5/5] END ...C=100, gamma=0.0001, kernel=rbf;, score=0.836 total time= 0.0s
[CV 1/5] END ...C=1000, gamma=1, kernel=rbf;, score=0.921 total time= 0.1s
[CV 2/5] END ...C=1000, gamma=1, kernel=rbf;, score=0.907 total time= 0.1s
[CV 3/5] END ...C=1000, gamma=1, kernel=rbf;, score=0.907 total time= 0.2s
[CV 4/5] END ...C=1000, gamma=1, kernel=rbf;, score=0.879 total time= 0.0s
[CV 5/5] END ...C=1000, gamma=1, kernel=rbf;, score=0.879 total time= 0.0s
[CV 1/5] END ...C=1000, gamma=0.1, kernel=rbf;, score=0.900 total time= 0.0s
[CV 2/5] END ...C=1000, gamma=0.1, kernel=rbf;, score=0.900 total time= 0.0s
[CV 3/5] END ...C=1000, gamma=0.1, kernel=rbf;, score=0.907 total time= 0.0s
[CV 4/5] END ...C=1000, gamma=0.1, kernel=rbf;, score=0.871 total time= 0.0s
[CV 5/5] END ...C=1000, gamma=0.1, kernel=rbf;, score=0.871 total time= 0.0s
[CV 1/5] END ...C=1000, gamma=0.01, kernel=rbf;, score=0.900 total time= 0.0s
[CV 2/5] END ...C=1000, gamma=0.01, kernel=rbf;, score=0.850 total time= 0.0s
[CV 3/5] END ...C=1000, gamma=0.01, kernel=rbf;, score=0.886 total time= 0.0s
[CV 4/5] END ...C=1000, gamma=0.01, kernel=rbf;, score=0.836 total time= 0.0s
[CV 5/5] END ...C=1000, gamma=0.01, kernel=rbf;, score=0.843 total time= 0.0s
[CV 1/5] END ...C=1000, gamma=0.001, kernel=rbf;, score=0.879 total time= 0.0s
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[CV 4/5] END ..C=1000, gamma=0.0001, kernel=rbf;, score=0.879 total time= 0.0s
[CV 5/5] END ..C=1000, gamma=0.0001, kernel=rbf;, score=0.836 total time= 0.0s

```

```

[19]: GridSearchCV(cv=5, estimator=SVC(),
                  param_grid={'C': [0.1, 1, 10, 100, 1000],
                              'gamma': [1, 0.1, 0.01, 0.001, 0.0001],
                              'kernel': ['rbf']}),

```

```
verbose=3)
```

```
[20]: grid.best_params_
```

```
[20]: {'C': 1000, 'gamma': 1, 'kernel': 'rbf'}
```

```
[21]: grid.best_score_
```

```
[21]: 0.8985714285714286
```

```
[22]: y_pred = grid.predict(X_new_test)
      print(accuracy_score(y_new_test, y_pred))
      print(classification_report(y_pred, y_new_test))
```

```
0.9166666666666666
```

	precision	recall	f1-score	support
0	0.98	0.87	0.92	167
1	0.86	0.98	0.91	133
accuracy			0.92	300
macro avg	0.92	0.92	0.92	300
weighted avg	0.92	0.92	0.92	300

So here using the Hyperparameter tuning we have increased the accuracy